

Homework 1 (20pts)

Why are we doing this? The homework assignments are a low-stakes way to practice the concepts we've talked about in class and apply them to real-world problems. For this particular assignment, I want you to think through the pros and cons of representing real-world systems as networks, get your hands dirty with some real network datasets, and visualize networks and measure their basic properties.

How will you be assessed? I am looking for critical thinking and well-thought-out responses. For the first problem, this involves picking systems which may have several plausible notions of “nodes” and “edges”, building upon our class activity. I'm hoping that you can express the pros and cons of each choice and practical ways to collect these networks. For the second problem, most of it is data science oriented and the labs should provide a good starting point for the code necessary to do this. The last question is a gentle introduction to the mathematics of networks.

What will you submit? A PDF of the homework assignment and all of the code (Jupyter notebooks or scripts) used to generate the results and figures to the Canvas assignment.

Format requirements

- The title of the homework assignment
- Your name
- The collaborators with whom you worked
- Each problem should have a separate heading
- Plots should have descriptive axis labels

The assignment

Extra credit (1 point): Use the LaTeX template (See Canvas) to write up the homework.

Question 1 (8 points): Choose **one** of the following systems:

- A social network among members of a dorm, club, class, major, or friend group based on your own knowledge or observations
- A co-occurrence network of organisms observed during a 10–15 minute walk outdoors
- A similarity network between individuals based on shared interests, tastes, courses, hobbies, or other characteristics collected by asking them.
- Another small network of your choice (please briefly describe it)

Collect enough information to construct a network with approximately 10-20 nodes. *My goal here is for this network to be small enough to be manageable, but large enough so that you see some interesting structure.* For the social network option, feel free to use abbreviations or pseudonyms to deidentify the data.

- a) (2 points) Write down your observations in an organized format (table, edge list, etc.)
- b) (1 point) Define what your nodes and edges represent. Explain why these choices make sense for the system you studied.
- c) (2 points) Draw or visualize your network. The visualization may be hand-drawn or generated

using software.

- d) (1 point) Describe one alternative choice of nodes or edges for the same system. How would this change the resulting network?
- e) (1 point) What type of network is most appropriate for your system? Should it be directed or undirected? Should it include edge weights, signs, self-loops, or multi-edges? Explain your reasoning.
- f) (1 point) Briefly describe how you collected the data. What sources of measurement error or bias might affect your network?

Question 2 (8 points): Download the behavioral beetle dataset CookSocial2020.

- a) Read the README.
- b) Clean the dataset to (1) remove beetles with unknown identities, (2) remove null values, and (3) restrict to 2020 observations with the following code:

```
import polars as pl
df = pl.read_csv("cook_social_2020.csv")
new_col = df["datetime"].str.to_datetime("%Y-%m-%dT%H:%M:%S%Z", strict=False)
df.replace_column(0, new_col)
df = df.drop_nulls()
```

```
bad_values = {'UK', 'UKM', 'UKF'}
```

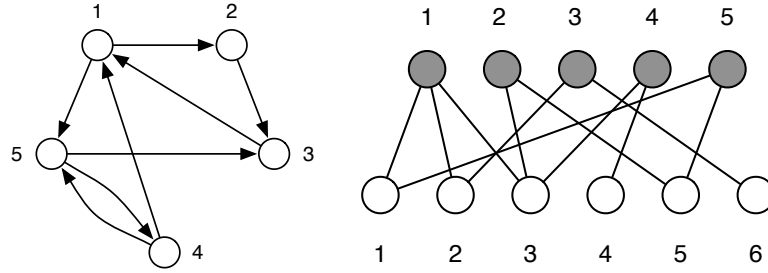
```
df = df.filter(~pl.col("focal_id").is_in(bad_values))
df = df.filter(~pl.col("partner_id").is_in(bad_values))
```

```
df = df.filter(pl.col("datetime").dt.year() == 2020)
```

```
condo_names = df["condo"].unique().to_list()
```

- c) (4 points) For the following, construct a dictionary of networks, where the keys are condo names and the values are NetworkX networks for the following networks:
 - i) The 5cm proximity network
 - ii) The touch network
 - iii) The mating network
- d) (2 points) Analyze the differences in behavior by:
 - i) A [strip plot](#), where the horizontal axis indicates the network type, and each point indicates the mean degree of the network of a single condo. Plot the average across all condos for each network type on the same plot.
 - ii) Explain the differences in mean in terms of behavior.
- e) (2 points) For one condo of your choice, plot the proximity, touch, and mating networks side-by-side. Briefly describe one difference you observe.

Question 3 (4 points): Consider the following two networks:



The first network is a *directed network* where links have direction and the second network is a *bipartite network* where there are two types of nodes and nodes of the same type cannot connect to one another.

- (2 points) Write out each network's edge list (hint: on network 2, it's easiest to re-index the grey nodes so their labels are 7 and higher) and use this to make a [spy](#) plot of its adjacency matrix.
- (1 point) For the first network, suppose that you know the sum of all out-degrees and the sum of all in-degrees except a single node, and that $\sum k_{in} - \sum k_{out} = \delta k > 0$. What do you know about the in-degree and out-degree of that node?
- (1 point) For the second network, derive a **general** expression relating the mean degree of the nodes of the first type (grey nodes) to the mean degree of the second type (white nodes). (General in the sense that it should hold for any bipartite network, not just the network shown.)
- (Extra credit: 1 point) Describe a possible extension of the clustering coefficient to directed networks (first network) and bipartite networks (second network). *There is not a right answer! Just use your creativity and justify why it makes sense.*